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Leveraging generative AI for course learning outcome categorization using Bloom's taxonomy

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ABSTRACT

Learning outcomes are clear and concise statements that describe what students should be able to do or know at the end of a particular course. These statements are crucial in instructional planning, curriculum development, and assessment of student progress and learning. Although there is no universal guidance on how to develop learning outcomes, Bloom's taxonomy is one widely used framework that helps instructors develop outcomes that reflect different levels of thinking, from basic remembering to creative problem-solving. This study investigates the potential of generative AI, specifically GPT-4, in classifying course learning outcomes according to their respective cognitive levels within the revised Bloom's taxonomy. To assess the effectiveness of GenAI, we conducted a comparative study using a dataset of 1000 annotated learning outcomes. We tested multiple prompt engineering strategies, including zero-shot, few-shot, chain-of-thought, rhetorical situation, and multiple binary questions, leveraging GPT-4. Classification performance was evaluated using accuracy, Cohen's κ, and F1-score. The results indicate that the prompt incorporating rhetorical context and domain-specific knowledge achieved the highest classification performance, while the multiple binary question approach underperformed even compared to the zero-shot method. Furthermore, we compared the best-performing prompting strategy with a state-of-the-art classification model, BERT. Although the fine-tuned BERT model showed superior performance, prompt-based classification exhibited moderate to substantial agreement with expert annotations. Overall, this article demonstrates the potential of leveraging large language models to advance both theoretical understanding and practical application within the field of education and natural language processing.

1. Introduction

Course learning outcomes (CLOs) communicate the instructional goals and serve as the basis for course design. These outcomes are expressed through clear statements that outline what students are expected to know and be able to do upon completing the course. Instructors use these outcomes to develop course activities and assessment items. Furthermore, using CLOs ensures consistency when different instructors teach the course and provides valuable insights into how the course aligns with broader institutional and accreditation standards (Orr et al., 2022).

Among various cognitive classification frameworks, the revised Bloom's taxonomy (Anderson et al., 2001; Bloom, 1956) stands out as the most widely used framework in curriculum development and instructional design. Educators and researchers frequently apply it to establish course learning outcomes, prepare educational materials, and assess exam questions (Shaikh et al., 2021). Additionally, accreditation organizations like ABET, the University Grants Commission (UGC), and the National Assessment and Accreditation Council (NAAC) have endorsed the revised Bloom's taxonomy as the primary framework for teaching, learning, and evaluation since it serves as a useful reference for curriculum development, creating assessment questions, and facilitating the measurement of intended outcomes (Sarkar, 2023). Its hierarchical structure, which moves from foundational skills to higher-order thinking skills-remember, understand, apply, analyze, evaluate, and create-provides a systematic approach to defining the complexity of learning outcomes (Anderson et al., 2001). These cognitive levels serve as labels for aligning CLOs with Bloom's cognitive taxonomy, an established theoretical educational framework. Furthermore, its previous use in learning outcomes research (Li et al., 2022, pp. 530–537;

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Shaikh et al., 2021) makes it the most suitable framework for this study.

Despite the widespread acceptance of Bloom's taxonomy, manual classification is prone to subjectivity, inconsistencies, and time constraints, prompting the need for automated classification methods to streamline this process. Masapanta-Carrión and Velázquez-Iturbide (2018, pp. 441–446) found that instructors struggled the most with Bloom's taxonomy when they use it to classify learning outcomes or assessment tasks. Another challenge of this taxonomy is that some of the action verbs overlap in multiple taxonomy levels, causing ambiguity about the appropriate cognitive level (Das et al., 2022). To tackle these issues, there is a growing interest in automating CLO classification to enhance consistency among instructors, ensure alignment with instructional activities and assessment items, and increase efficiency.

Recent advancements in GenAI, particularly the development of Natural Language Processing (NLP) based large language models (LLMs) such as GPT-4, have shown significant effectiveness in classification tasks. Unlike traditional methods that require extensive labeled datasets and specialized technical skills, prompt engineering has emerged as a flexible and accessible alternative for optimizing model performance without the need for additional training. By strategically designing prompts, researchers and practitioners can enhance LLM classification accuracy by minimizing ambiguity, improving interpretability, and aligning model predictions with expert judgment (Brown et al., 2020; Wei et al., 2022).

In this study, we investigate the effectiveness of prompt engineering in automatically classifying course learning outcomes (CLOs) into cognitive levels based on the revised Bloom's taxonomy. We systematically evaluate various prompting strategies, including zero-shot, few-shot, chain-of-thought, rhetorical situation, and multiple binary question approaches using GPT-4. Additionally, we compare the performance of prompt-based classification against a fine-tuned BERT model on the same dataset. Through this analysis, we aim to provide practical insights into optimizing prompt-based methodologies for CLO classification, facilitating more efficient and consistent alignment of learning outcomes with Bloom's cognitive level taxonomy. This research contributes to the broader discourse on integrating AI-assisted methods in curriculum planning and educational assessment.

2. Related work

2.1. Prompt-based learning for text classification

Prompt engineering or design, i.e., the use of prompts, has become a common and increasingly popular mechanism for leveraging the power of a LLM. Generative Pre-trained Transformer (GPT) models have shown considerable promise across various classification tasks. It has outperformed conventional machine learning algorithms at various classification tasks, including sentiment analysis, news topic classification (Zhu et al., 2023), and political stance (Chae & Davidson, 2023).

Another widely used NLP model is Bidirectional Encoder Representations from Transformers (BERT), a transformer-based deep learning model designed to pre-train text representations in a bidirectional manner (Devlin et al., 2019). BERT has achieved state-of-the-art performance in numerous NLP tasks, including text classification, named entity recognition, and question-answering. Zhong et al. (2023) found that ChatGPT displayed comparable performance to BERT in certain natural understanding tasks, such as sentiment analysis and question-answering tasks.

On the other hand, Mayer et al. (2023) evaluated fine-tuned pre--trained models and different prompt-based learning approaches for text classification. Fine-tuned BERT models refer to BERT models that have been further trained (fine-tuned) on specific datasets tailored to a particular task, optimizing performance beyond the original pre-training. Their study found that the fine-tuning pre-trained models outperformed zero and few shots learning, yet prompt-based learning is worth exploring since it has the potential to enable a wider audience to use LLMs without requiring advanced programming skills or the need for extensive fine-tuning of models with large amounts of labeled training data. Wang et al. (2024) found that teachers exhibited greater trust and technology acceptance regarding ChatGPT compared to BERT in supporting their acquisition of dialogic pedagogy.

Although the use of prompting has made some aspects of NLP easier, it requires careful design, as prompting strategies influence the performance of the GPT models. Consequently, several advanced strategies have emerged to optimize model outputs, each with distinct mechanisms and advantages. One common approach is zero-shot prompting, where the model is given a task description but no examples. In this method, the model generates responses solely based on its pre-trained knowledge (Brown et al., 2020). While zero-shot prompting is efficient, its accuracy often suffers due to a lack of contextual grounding. In contrast, few-shot prompting, also known as in-context learning, provides the model with a few labeled examples within the prompt to guide it toward the desired output (Brown et al., 2020). Research suggests that few-shot learning generally improves classification performance compared to zero-shot prompting, but its effectiveness is highly sensitive to the choice and quality of examples (Chae & Davidson, 2023). Additionally, Chae & Davidson (2023) demonstrated that increasing the richness of contextual information within prompts leads to higher predictive accuracy, reinforcing the need for well-designed prompts.

Beyond standard in-context learning, researchers have explored chain-of-thought (CoT) prompting, which encourages the model to explicitly reason through intermediate steps before arriving at a final answer. By breaking down the problem into logical steps, CoT prompting has been shown to enhance performance in complex reasoning and classification tasks (Wei et al., 2022). A related extension is zero-shot CoT prompting, where simple instructions (e.g., "Let's think step by step") are provided without labeled examples, prompting the model to induce reasoning without the need for in-context examples (Kojima et al., 2022). This approach is particularly appealing as it improves the model's logical reasoning capabilities while maintaining efficiency.

A more recent strategy, rhetorical situation prompting, has been proposed to incorporate contextual and rhetorical elements within the prompt to better align the model's responses with user intent. Ranade et al. (2024) suggest that incorporating rhetorical framing allows for more user-friendly prompt engineering, particularly in education settings, where instructional context plays a significant role in the interpretation of learning objectives. This method is distinct from conventional prompting techniques in that it explicitly considers how the linguistic framing of prompts influences model responses, making it a promising avenue for educational NLP applications.

Most existing studies focus on general classification problems (e.g., sentiment analysis, topic modeling, and question answering), lacking empirical evaluations that compare these strategies in the context of educational NLP tasks, specifically, the cognitive level classification of course learning outcomes using Bloom's taxonomy. This study aims to address this research gap by empirically assessing the effectiveness of zero-shot, few-shot, CoT, zero-shot CoT, and rhetorical situation prompting for CLO classification. In doing so, we contribute to the ongoing discussion about how different prompt designs influence LLM performance in educational settings.

2.2. Automatic classification based on Bloom's taxonomy

Educational text classification based on Bloom's Taxonomy primarily focuses on categorizing Course Learning Outcomes (CLOs) and assessment items, such as exam questions, according to their respective levels of cognitive processes. Unlike general text classification tasks, which deal with sentiment analysis, topic modeling, or named entity recognition, classification based on Bloom's Taxonomy requires an understanding of hierarchical cognitive processes ranging from lower-order (e.g., remembering, understanding) to higher-order thinking (e.g., analyzing, evaluating, creating). This hierarchical nature introduces unique challenges, such as distinguishing between adjacent cognitive levels and addressing instances where a single CLO may span multiple levels.

There has been an evident shift in the approaches used in educational text classification based on (Revised) Bloom's Taxonomy in recent literature. Fully supervised machine learning studies have explored various techniques for feature extraction and selection, including term frequency (TF), POS-based weighting, and document distribution by class category which have been found to be significant (Gani et al., 2022). Other researchers have also investigated combining feature extraction techniques with embeddings, for instance, employing TFPOS-IDF + word2vec (Mohammed & Omar, 2020) and TFPOS-IDF + GloVe (Modi et al., 2022, pp. 25–37). As for machine learning techniques for model building, support vector machine and logistic regression have emerged as effective methods, depending on the task, dataset, and feature extraction technique used (Ahmed & Sorour, 2024; Modi et al., 2022, pp. 25–37; Mohammed & Omar, 2020). These methods require complex feature engineering and often lack scalability.

The emergence of pre-trained models has addressed these limitations by enabling implicit feature representation. Researchers have assessed different word embedding techniques, such as GloVe, FastText, BERT, and RoBERTa, with different deep learning techniques, such as CNN, LSTM, and fine-tuned BERT (Banujan et al., 2023; Das et al., 2020; Gani et al., 2023; Laddha et al., 2021; Li et al., 2022, pp. 530–537; Shaikh et al., 2021; Sharma et al., 2023; Zhang et al., 2021). Banujan et al. (2023) found that BERT + LSTM outperformed GloVe + LSTM and TF-IDF + ANN. Gani et al. (2023) found that RoBERTa using CNN outperformed other word-embedding techniques including word2vec, FastText, GloVe, and BERT. In a comparative analysis for cognitive level classification based on Bloom's taxonomy, machine learning techniques have been outperformed by LSTM (Shaikh et al., 2021) and fine-tuned BERT models (Li et al., 2022, pp. 530–537; Sebbaq & Faddouli, 2022). The performance of these models heavily depends on the quality and quantity of the training set and thus struggles to identify cognitive levels that do not have enough representation in the training dataset.

Most prior work assumed the problem was a single-label classification task, which is not always the case. In a large learning outcome dataset collected from thousands of courses, more than 10 % of the CLOs belong to more than one cognitive level (Li et al., 2022, pp. 530–537). Li et al. (2022, pp. 530–537) have compared the performance of machine learning and deep learning methods to automatically classify learning outcomes into the appropriate cognitive levels of Bloom's taxonomy. They have also tested two approaches, which are the use of multiple binary classifiers for each cognitive level, and the use of one multi-class multi-label classifier for all cognitive levels. The results showed that BERT outperformed other conventional machine learning methods (namely, NV, SVM, LR, RF, and XGBoost). In addition, using multiple binary classifiers (one binary classifier for each cognitive level) achieved better performance than the multi-label multi-class classifier.

Building on previous research, the current study examines the effectiveness of prompt-based learning in classifying CLO into its cognitive levels and compares this with BERT. Additionally, it investigates whether using multiple questions yields more accurate classification results, as observed in fully supervised machine learning and deep learning techniques (Li et al., 2022, pp. 530–537).

3. Research contribution

Prompt-based learning has been selected for evaluation due to its demonstrated efficacy in similar tasks, indicating promising potential for addressing the challenge of automatically classifying course learning outcomes within Bloom's Taxonomy. In addition, the use of prompting offers a compelling advantage for educators and practitioners, as it relies on natural language instruction, eliminating the need for technical expertise. However, the effectiveness of different prompting strategies for course learning outcomes classification remains underexplored, necessitating further investigation. To bridge this gap, this study seeks to answer the following three research questions:

RQ1. Which prompt strategy is the most effective for classifying CLOs into Bloom's cognitive levels? Prior research has demonstrated that different prompting strategies can yield varying levels of effectiveness in text classification tasks. However, their comparative performance in the context of classifying course learning outcomes within Bloom's taxonomy remains unclear. The findings will help formulate practical recommendations for enhancing collaboration between educators and LLMs in CLO classification.

RQ2. Can the utilization of multiple questions, each tailored to a specific cognitive level, akin to binary classifiers, yield improved outcomes in prompt-based learning? Traditional text classification often benefits from multi-class binary classifiers trained on specific categories. By structuring prompts in a similar manner, we investigate whether such an approach would enhance the model's ability to capture nuanced conceptual distinctions. The results will deepen our understanding of NLP applications in education.

RQ3. Does prompt-based learning outperform the pre-trained, fine-tuned BERT model in classifying CLOs into Bloom's cognitive levels? The literature presents mixed results regarding the effectiveness of GPT-based models in text classification compared to fine-tuned transformer models like BERT. This study directly compares these approaches to determine whether prompt-based learning can match or surpass the performance of fine-tuned models in CLO classification. The findings will offer insights into the viability of prompt-based approaches as a lower-barrier alternative for educators.

Overall, this paper contributes to addressing some of the existing challenges educators face with generative AI. These challenges include a lack of understanding of how to utilize GenAI to improve teaching efficiency, as well as a lack of awareness of the educational benefits offered by generative AI (Lan & Chen, 2024). As Kizilcec et al. (2024) pointed out, integrating LLMs into curricula and assessments is not an innovation but a necessity to align learning outcomes with real-world challenges in the labor market. Thus, by examining prompt engineering strategies, segmentation techniques, and model performance comparisons, we aim to advance both theoretical understanding and practical applications within the field of educational technology and NLP.

4. Methodology

Fig. 1 illustrates the workflow of this research, which aims to address the three research questions posed. The dataset is divided into training and testing sets. The training set will be used to fine-tune BERT and develop a classification model, while the testing set will be employed to evaluate BERT and various prompt engineering strategies.

4.1. Data

In this study, we utilized the dataset from Li et al. (2022, pp. 530–537), comprising 21,380 labeled learning objectives from 5558 courses across 10 faculties at an Australian university in 2021. The dataset was primarily annotated by a trained individual knowledgeable in Bloom's taxonomy. To enhance reliability, a second annotator reviewed 30 % of the dataset, and inter-rater reliability was assessed using Cohen's kappa, resulting in a value of 0.63. After resolving discrepancies, the inter-rater agreement for the same sample improved to 0.80, establishing it as a reliable benchmark for course learning outcomes classification.

Table 1 presents examples of learning outcomes alongside their corresponding cognitive level. The majority of the learning objectives were assigned to a single cognitive level, with around 12 % being classified to more than one cognitive level. In particular, 2325 learning objectives were associated with two cognitive levels. Consequently, this was considered a multi-label classification task.

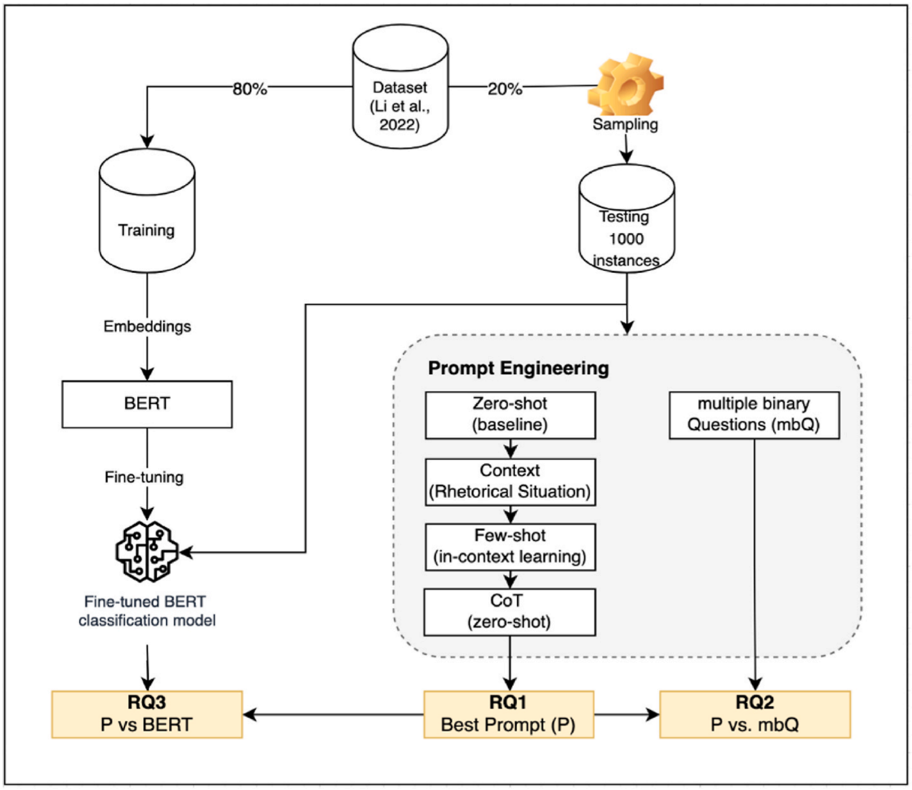

Fig. 1. Research workflow diagram.

Table 1
Examples of CLOs and their associated cognitive level.

#	Course Learning Outcome	Cognitive Level(s)
1	Investigate foods, food ingredients and varied cultural eating patterns found in Australia.	Analyze
2	Show technical skill in designing, conducting and reporting on a research project.	Apply
3	Identify and interpret the means by which the child communicates in therapy.	Understand & Evaluate

The dataset was split into two corpora - 80 % for training and fine-tuning BERT, and 20 % for testing. Due to the associated cost of using OpenAI API, 1000 instances were randomly selected from the testing set for evaluation. Fig. 2 shows the distribution of the testing dataset (a) across different cognitive levels, and (b) according to the number of cognitive level(s) associated with a CLO. The distribution across cognitive levels is not uniform because the dataset reflects the natural variability of cognitive complexity in the source material. Certain cognitive levels are inherently more frequent in educational content, leading to an uneven distribution that maintains the authenticity of the dataset.

4.2. Prompt engineering

The process of creating prompts that result in the most effective performance to solve the task at hand is called prompt engineering (Liu et al., 2023). In this study, we evaluated several prompt strategies, each executed using GPT-4. After assessing all strategies, a combination of the top-performing approaches was selected as prompt (P). These strategies differ in their mechanisms for guiding LLMs in classification tasks, each addressing distinct challenges by applying varying levels of guidance, supervision, and task decomposition. This structured approach enables a comprehensive evaluation of LLMs' ability to classify CLOs across cognitive levels. Next, we provide a description of the strategies and their implementation in this study.

4.2.1. Zero-shot learning

This is the baseline for prompt-based learning, as no examplesare provided for the task of interest. The prompt is a direct request to classify a given CLO without assigning a role, context, or examples. The result presents the performance of the underlying LLM in classifying CLOs into cognitive levels.

4.2.2. Context using Rhetorical Situation

Ranade et al. (2024) suggested the rhetorical situation as a framework to assist writers in developing reliable and efficient prompts. As a result, they created a formula consisting of the following elements: (audience + genre + purpose + subject + context + exigence + writer). These components can be modified based on the context and the desired output. We adopted their strategy as it gives a rich context to the prompt, as shown in Table 2. While this framework informs the design to ensure that key contextual elements are included, the prompt-writing process itself remains manual.

4.2.3. Few-shot (in-context) learning

Few-shot prompting, as outlined by Brown et al. (2020), is a method used to direct the model toward desired behavior by providing labeled instances as examples. A prompt typically consists of a tuple of <input, output>, concluding with an input designed to prompt the model to complete the cloze with the output. Although the approach has proven effective across various NLP tasks, researchers found that the performance of few-shot (in-context) learning is susceptible to the number and content of the provided examples (Chae & Davidson, 2023; Zhong et al., 2023). In this study, we adjust the technique by providing a definition and five examples of common action verbs used at each cognitive level. The snippet below shows an example of the 'Remember' cognitive level. The aim is to provide examples and context without improperly influencing the model with subject-specific terminology.

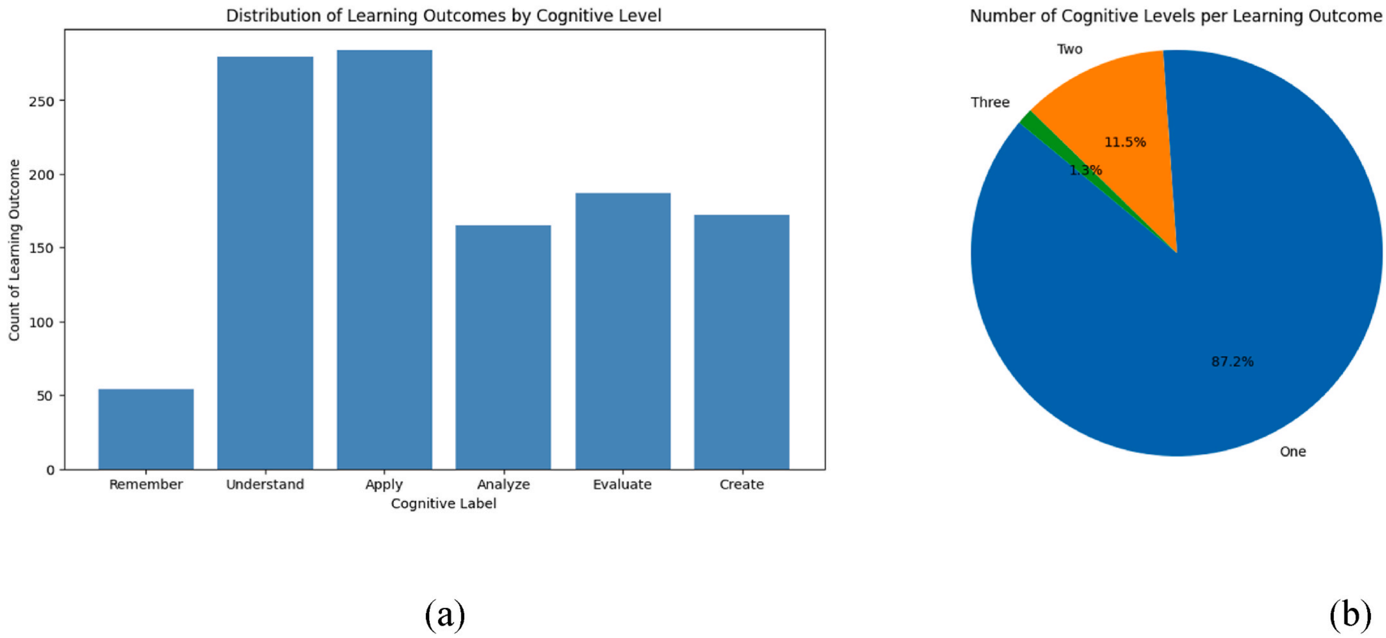

Fig. 2. The distribution of testing dataset (a) across cognitive levels (b) number of cognitive levels per CLO.

Table 2
Rhetorical situation elements for prompt writing.

Element	Definition (Ranade et al., 2024)	Content in the prompt
Genre	Preferred selected genre	course syllabus
Audience	The intended reader demographic of the content	instructor
Purpose	The author's/reader's purpose for writing/ reading	assigning a cognitive level for each CLO based on the revised Bloom's Taxonomy.
Subject	Focus and details of the content	categorize the CLO below … Note that a CLO may belong to one or more of the following six cognitive levels: remember, understand, apply, analyze, evaluate, and create.
Context	Relevant circumstances outside of the content	according to the appropriate cognitive level from the revised Bloom's Taxonomy. Note that a CLO may belong to one or more of the following six cognitive levels: remember, understand, apply, analyze, evaluate, and create
Exigence	The demand to write the content	needs assistance
Writer	The intended writer of the content	an experienced course designer

4.2.4. *Chain-of-thought*

Wei et al. (2022) introduced the chain-of-thought (CoT) prompting strategy, which uses triples: <input, chain of thought, output>, where a chain of thought represents a series of intermediate natural language reasoning steps leading to the final output. CoT, as described by Wei et al. (2022), applied few-shot learning, which requires careful human engineering of prompt examples. In contrast, Kojima et al. (2022) proposed zero-shot-CoT, which requires less engineering by simply appending 'Let's think step by step.' to the end of the prompt. Despite underperforming relative to few-shot-CoT, zero-shot-CoT outperformed the zero-shot approach and produced a plausible reasoning path (Kojima et al., 2022). In this study, we utilized zero-shot-CoT for its practicality.

4.2.5. *Multiple binary questions (mbQ)*

In this strategy, the prompt is formulated to encompass six binary questions, each targeting a specific cognitive level. The question is formulated as: "Should this CLO be categorized under the 'X' cognitive level? (Yes/No)", where X {'remember', 'understand', 'apply', 'analyze', 'evaluate', 'create'}.

In summary, Zero-shot learning represents the most fundamental but often the least robust method, primarily used to evaluate the model's innate capabilities without external guidance. Context-based prompting, utilizing the Rhetorical Situation, enhances the contextual richness of the prompt rather than providing explicit labeled examples, offering a structured yet flexible approach to guiding the model. In contrast, few-shot learning introduces specific examples, providing a form of implicit training that can improve classification performance. The Chain-of-Thought (CoT) methodology extends the model's reasoning ability by structuring its thought process step by step. Similarly, the multiple binary Questions (mbQ) strategy decomposes the classification task into discrete binary decisions, which can improve accuracy, particularly in multi-label classification scenarios. The study integrates these strategies to optimize the overall performance. The utilized prompts are listed in Appendix 1.

4.3. *Models*

GPT-4 was chosen for its state-of-the-art performance in generative AI, while BERT was selected due to its strong track record in educational NLP tasks, particularly text classification. The fine-tuned BERT model and the various prompts using GPT-4 were assessed using the same dataset. This direct comparison provides deeper insights into the trade-offs between flexibility and accuracy in CLO classification.

4.3.1. *Pre-train, prompt approach: GPT-4-turbo*

GPT-4 is a large multimodal model that achieved human-level

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Remember
Definition: retrieve, recall, or recognize relevant knowledge from long-term memory.
Examples of learning outcome verbs: list, recall, state, name, and define.
```

performance on several academic benchmarks. The model is trained up to April 2023, ensuring it has the latest knowledge and insights, with a 128,000-token context window, allowing it to accommodate longer text and more comprehensive interactions in a single prompt. The latest version, GPT-4-turbo, enhanced instruction following, JSON mode, reproducible outputs. The utilized version of the model is 'gpt-4-turbo'. Regarding the model configuration, the temperature was set to zero to make sure we get the most probable sequence of tokens for the classification task and to minimize variation in the results. All other parameters were maintained at their default settings.

4.3.2. Pre-train, fine-tune approach: BERT

Adopting the multilabel multiclass model architecture in (Li et al., 2022, pp. 530–537), where the base model is BERT-uncased. The model included 12 hidden layers, each with 768 neurons, and the dropout rate was 0.1. The entire BERT model was fine-tuned without freezing parameters in any layer. The batch size was set to 64, and the number of epochs was set to 3. Early-stopping was utilized to avoid overfitting using 20 % of the 80 % of training data for validation. When the F1 scores stopped improving on five consecutive validations, the training terminated, and the model weights rolled back to the best-performing one.

4.4. Evaluation

The prompts variation, as well as the fine-tuned BERT model, were evaluated using the known classification evaluation metrics: accuracy, Cohen's κ, and F1 score. Accuracy measures the proportion of correctly classified course learning outcomes, while Cohen's κ evaluates inter-rater agreement, accounting for chance agreement. The F1 score effectively balances precision and recall. Collectively, these metrics provide a comprehensive evaluation of model accuracy and alignment with expert assessments of course learning outcomes classification. The classification results for each cognitive level were separated to assess their performance individually.

5. Results

In this study, we initially evaluated individual prompts and then experimented with combinations to determine whether they could improve the results. We retained the combinations P2+P3 and P2+P3+P4 because they led to performance enhancements while discarding those that did not lead to better outcomes. The rationale behind combining different strategies is that they could complement one another and thereby improve classification accuracy. For full transparency, the complete prompt text used in each strategy is provided in Appendix A.

RQ1. Which prompt strategy is the most effective for classifying CLOs into Bloom's cognitive levels?

Table 3 shows the number of cognitive levels identified for each course learning outcome (CLO) across the various prompts. P1 and P2 predominantly identified a single label (cognitive level) for each CLO. Although an explicit note was stated in the prompt that a "CLO may belong to one or more of the following six cognitive levels: remember, understand, apply, analyze, evaluate, and create.", no CLO was assigned more than one label. Furthermore, there were some cases in P1-P3 where no appropriate cognitive level was assigned to a CLO. Notably, the utilization of CoT in P4 has facilitated the assignment of cognitive levels to all CLOs. At the same time, it seems to overanalyze the CLOs assigning 3, 4, and 5 labels to a CLO.

Table 4 provides a detailed comparison of classification performance across different prompts using three key evaluation metrics: accuracy, Cohen's κ, and F1-score. The results suggest that the initial prompt P1 is promising. In essence, posing a straightforward question showed a moderate to substantial agreement with expert annotation across the six cognitive levels per Landis and Koch (1977) interpretation of Cohen's κ scores.

P2 had more specifications in the prompt, such as the role, purpose, and context, which aimed to provide context and set the overarching goal of the task. Although it could not label the appropriate cognitive level for a larger number of CLOs, it has outperformed the baseline P1 in three cognitive levels. A similar observation was noticed in the result of P3, with even more instances (~7 %) not categorized to any cognitive level. Nevertheless, the performance of P3 was also better than the baseline in four out of the six cognitive levels. Combining P2 and P3 (P2+P3) produced even better results. The combined approach (P2+P3) not only resulted in fewer CLOs without an assigned cognitive level, as shown in Table 3, but also demonstrated superior classification performance across multiple Bloom's cognitive levels. Specifically, it improved classification accuracy in the lower-order levels (Remember, Understand) as well as in higher-order levels (Analyze, Create). This suggests that incorporating both contextual information (P2) and in-context examples (P3) enhances the model's ability to recognize and classify CLOs across both foundational and more complex cognitive skills.

P4 is a chain-of-thought prompt. The performance metrics did not show improvement in the classification results over the baseline. Combining P2, P3, and P4, however, has led to improved performance in three cognitive levels compared to the P2+P3. Yet, the increased performance is offset by a larger loss in the other three categories. This suggests that the superior model remains P2+P3. The implementation of this prompting strategy has notably enhanced Cohen's κ, which initially ranged from 0.484 to 0.684 in P1 to 0.555–0.720 in P2+P3.

RQ2. Can the utilization of multiple questions, each tailored to a specific cognitive level, akin to binary classifiers, yield improved outcomes in prompt-based learning?

Table 5 shows the classification performance results of the multiple binary questions (mbQ) strategy. Although this approach aimed to replicate the effectiveness of binary classifiers found in traditional machine learning, it did not surpass the baseline prompt (P1), with the sole improvement observed in the 'Create' category. This suggests that creating multiple binary questions may result in an overgeneration of labels and greater confusion, ultimately reducing overall classification accuracy.

RQ3. Does prompt-based learning outperform the pre-trained, fine-tuned BERT model in classifying CLOs into Bloom's cognitive levels?

Table 6 presents the classification performance results of the fine-tuned BERT model across the six cognitive levels. As shown in the table, fine-tuned BERT consistently outperformed all tested GPT-4 prompting strategies, achieving the highest accuracy, Cohen's κ, and F1-score across all cognitive categories. The model demonstrated near-perfect agreement with human labels (Cohen's $\kappa > 0.84$).

6. Discussion

The analysis of the effectiveness of using prompts-based learning to classify learning outcomes into cognitive levels showed that a simple (zero-shot) prompt has yielded moderate to substantial agreement with experts, with a minimum accuracy of 0.857 and an F-1 score of 0.5. The best prompt has been shown to be a combination of different strategies,

Table 3
The number of labels (cognitive levels) associated with each CLO in the various prompts.

Number of Labels	P1	P2	P3	P2+P3	P4	P2+P3+P4
0	9	22	66	3		
1	991	978	915	992	852	815
2			18	4	113	147
3			1	1	25	34
≥4					10	4

*P1: zero-shot prompt, P2: adding context using rhetorical situation, P3: few-shot (in-context) prompt, P4: chain-of-though (zero-shot) prompt.

Table 4
Performance classification metrics (accuracy, Cohen's κ, and F1-score) for different prompting strategies across the six cognitive levels of Bloom's taxonomy.

	Remember			Understand			Apply		
	Acc.	κ	F1	Acc.	K	F1	Acc.	κ	F1
P1	0.962	0.484	0.500	0.883	0.685	0.760	0.860	0.642	0.737
P2	0.954	0.375	0.395	0.887	0.708	0.784	0.861	0.634	0.725
P3	0.954	0.375	0.395	0.890	0.719	0.794	**0.887**	**0.708**	**0.783**
P2+P3	**0.964**	**0.573**	**0.591**	**0.891**	**0.720**	**0.793**	0.866	0.661	0.753
P4	0.949	0.347	0.370	0.847	0.599	0.701	0.849	0.620	0.724
P2+P3+P4	0.957	0.473	0.494	0.875	0.682	0.767	0.869	0.676	0.767
	Analyze			Evaluate			Create		
	Acc.	κ	F1	Acc.	K	F1	Acc.	κ	F1
P1	0.862	0.531	0.615	0.857	0.558	0.647	0.890	0.540	0.599
P2	0.862	0.533	0.617	0.856	0.530	0.619	0.893	0.556	0.614
P3	**0.883**	0.564	0.633	**0.877**	**0.593**	**0.668**	0.888	0.531	0.591
P2+P3	**0.883**	**0.583**	**0.653**	0.866	0.556	0.638	**0.892**	0.555	0.614
P4	0.795	0.421	0.541	0.844	0.553	0.650	**0.892**	**0.582**	**0.645**
P2+P3+P4	0.846	0.520	0.613	0.845	0.557	0.653	0.887	0.566	0.632

Table 5
Performance metrics for mbQ (P5) at the different cognitive levels.

	Acc.	Cohen's κ	F1-score
Remember	0.957	0.560	0.583
Understand	0.523	0.214	0.533
Apply	0.751	0.496	0.677
Analyze	0.722	0.374	0.522
Evaluate	0.756	0.423	0.567
Create	0.882	**0.621**	**0.693**

Table 6
Performance metrics for fine-tuned BERT at the different cognitive levels.

	Acc.	Cohen's κ	F1-score
Remember	**0.991**	**0.917**	**0.922**
Understand	**0.958**	**0.896**	**0.926**
Apply	**0.954**	**0.888**	**0.920**
Analyze	**0.967**	**0.876**	**0.895**
Evaluate	**0.976**	**0.921**	**0.936**
Create	**0.959**	**0.849**	**0.873**

particularly by including descriptive context, as demonstrated in P2+P3 and P2+P3+P4. These findings are consistent with prior research demonstrating that large language models exhibit strong performance in zero-shot and few-shot classification tasks; however, their accuracy is significantly influenced by the structure of the prompts used and the complexity of the tasks at hand (Brown et al., 2020; Zhong et al., 2023).

In addition, upon examining the responses for instances that have not been classified into any cognitive level, it was noticed that the model has responded by either repeating the CLO or extracting the first verb in the CLO, such as "identify" and "integrate", as a cognitive level. It was noteworthy to observe the increase in the quantity of CLOs that were not categorized into any cognitive level during P2 and P3. This posed a new question regarding the potential impact of providing longer context and whether it may lead to GPT's laziness.

This work also highlights other potential limitations of the effectiveness of using GenAI for educational tasks. For instance, another interesting finding is that using the CoT feature has not improved the accuracy of classification. This may be because GPT-4 tends to over-analyze the input, leading to multiple cognitive levels being assigned to a CLO, which was evident in Table 3 with an increased number of CLOs with more than one label. However, leveraging a chain of thought promoting strategy has the advantage of providing a rationale, which offers a level of transparency and enables humans to understand the machine's thought process behind classifying a CLO into a specific cognitive level.

In analyzing the mbQ strategy, the findings revealed that approximately 70 % of the CLOs fell into two or more cognitive categories. This suggests that asking multiple direct questions such as "Should this CLO be categorized under the [cognitive level]?" may have influenced the responses and prompted more thoughtful consideration for each level independently. While this strategy has negatively impacted the classification, it did improve agreement with experts in the "Create" cognitive level. Additionally, Contrary to the expectation that multi-class binary classifiers may enhance performance, as observed in Li et al. (2022, pp. 530–537), using multi-class binary questions for the GenAI did not lead to improved results.

Four examplesof CLOs labeled by experts and GPT-4, accompanied by GPT-4 rationale, using P2+P3+P4, are presented in Appendix 2. Upon analyzing these examples, concerns have arisen about the reliability of the dataset labeling. For example, in the first and second instances, the GPT labels and rationale are convincing. However, the last two examples highlight instances where the GPT label and rationale failed to capture the overall intent of the CLO. The errors in both experts and GPT-4 classification tend to be predominantly due to focusing on the first verb and disregarding the context.

Last but not least, the study has shown that fine-tuning BERT model was superior, achieving almost perfect agreement with expert labels. This is consistent with findings that fine-tuned BERT models generally outperform prompt-based approaches in domain-specific classification tasks (Mayer et al., 2023). In addition, the success of the fine-tuned model could be attributed to training it on a labeled dataset, and then in the evaluation, it was tested on an unseen dataset yet labeled by the same expert. Therefore, it is important to mention that the generalizability of the fine-tuned BERT model has not been verified, as it may have absorbed the annotator's biases (i.e., subjective interpretation).

7. Conclusion

This study has many implications that advance both theoretical understanding and practical implementation within the field of educational technology and NLP. It has carried out multiple experiments showing the effectiveness of GenAI and the utilization of LLMs to categorize course learning outcomes into cognitive levels through prompts. Although fine-tuning pre-trained models such as BERT is superior in terms of the accuracy of this classification task, the prompt-based approach has also yielded promising results. This suggests an opportunity for further exploration of prompt-based learning methods to maximize the potential of large language models.

From a practical perspective, this research highlights how prompting LLM models can be utilized for curriculum alignment by automatically classifying CLOs with minimal manual effort. Educators can incorporate accessible GenAI, such as ChatGPT, models into curriculum development to ensure consistency and enhance course design efficiency. Additionally, these models can improve assessment design by efficiently aligning questions with the intended cognitive levels. By streamlining these processes, generative AI tools can offer a scalable and impactful solution for integrating AI into educational practice.

Moreover, the analysis of various prompting strategies highlighted elements that enhanced the basic prompts and led to improved results. The optimal prompt encompassed a detailed description utilizing two strategies: the rhetorical situations, which serves to set up the context and the overarching goal of the task, and the knowledge domain context, which provides definitions and examplesof the categories of the theoretical framework; in this case, cognitive levels based on the revised Bloom's taxonomy. The chain-of-thought approach adds a level of transparency that educators may be interested in, although it did not improve the accuracy during this study. Future research may look closer into this particular strategy to pinpoint the issue or improve the prompt.

Additionally, this research sheds light and provides empirical findings on the mbQ strategy in prompt engineering. While classical machine learning approaches have extensively studied multiple binary classifiers versus multilabel multiclass classifier, the effectiveness of mbQ has received less attention. This finding provides valuable insight by demonstrating that mbQ has not resulted in better performance, as is the case with multiple binary classifiers used in classical machine learning. On the contrary, posing a single general question has produced better results.

Although the dataset used in this study originates from Australian universities, its broad coverage across multiple disciplines enhances the generalizability of the findings to a wider range of subject areas. Furthermore, the use of widely recognized educational frameworks, such as Bloom's taxonomy, reinforces the applicability of the methodology beyond this specific context. Future research could explore extending this approach to the classification of assessment items within Bloom's taxonomy or adapting it to alternative educational frameworks to further assess its robustness and versatility.

Finally, we are aware of the many potential ethical concerns with using GenAI within education (Holmes & Miao, 2023). To limit potential privacy concerns, we limited the study to one dataset that was developed and shared for research in keeping with responsible research guidelines. The dataset was also valuable because it includes a large number of labeled CLOs from different universities, encompassing a wide variety of subjects. Future research should address concerns with responsible development and use of datasets further, develop other datasets to increase generalization and broaden the scope beyond CLO classification by evaluating the classification on assessment questions. *This step would help bridge the gap between CLO classification and direct instructional applications, such as AI-assisted grading and feedback mechanisms.* Overall, for educational research it is critical to build a responsible research infrastructure for research related to AI and GenAI (Porayska-Pomsta, 2024). Additionally, future research could explore the classification of CLOs based on the program's outcomes or consider alternative theoretical frameworks. This research contributes to understanding how to utilize GenAI to improve teaching efficiency and advance both theoretical understanding and practical implementation within the field of educational technology and NLP.

CRediT authorship contribution statement

Omaima Almatrafi: Writing – original draft, Formal analysis, Conceptualization. **Aditya Johri:** Writing – review & editing, Funding acquisition.

8. Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used Grammarly in order to improve language and readability. After using this tool/service, the author(s) reviewed and edited the entire manuscript and take(s) full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.caeai.2025.100404.

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